



Indirect Energy in LCA - Shared Methodology (Knitting) - Modelling Approach, Calculation Framework and Default Values

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🕒 Created time	April 21, 2026 7:41 PM
☰ Category	Methodology doc
👤 Last edited by	🕒 Gustavo Vieira
🕒 Last updated time	April 23, 2026 12:08 AM

Overview

This page is the **single sourced home** for the indirect energy methodology used across all Carbonfact knitting process datasets (flat, circular, seamless, 3D, hosiery). Each process doc links here instead of repeating the derivation.

Default value adopted: 4.46 kWh/kg fabric (temporary reference pending sub-metered primary data).

1. Rationale

Knitting plants consume electricity across multiple subsystems beyond the knitting machines themselves. Sub-metered data are rarely available in the

literature; most published studies report only total plant electricity. To maintain transparency and consistency, the indirect component is modelled as a single aggregate value derived from literature-based subsystem shares.

This value is **identical across all knitting technologies** (flat, circular, seamless, 3D, hosiery) until factory-specific sub-metered measurements are available.

2. Subsystems Included

Subsystem	Share (%)	kWh/kg	Rationale
Knitting machines	35	1.56	Main mechanical load
Compressors	25	1.11	Pneumatic needle actuation
HVAC	30	1.34	Air conditioning and humidification
Lighting	6.5	0.29	Facility lighting
Cleaning / auxiliaries	3.5	0.16	Maintenance and automation units
Total	100	4.46	

3. Derivation of 4.46 kWh/kg

The distribution above is calibrated primarily on the weaving benchmark from Koç & Çinçik (2010), which is the closest published technology with a full subsystem breakdown. The knitting machine share (35%) is consistent with the weaving analogy and supported by the sources below.

Starting from the **upper EcoBalyse estimate for knitting machine electricity** of 2.4 kWh/kg (ADEME & Ministère de la Transition Écologique, 2024), representing 35% of total plant electricity:

$$E_{\text{total}} = \frac{2.4}{0.35} = 6.86 \text{ kWh/kg}$$

$$E_{\text{indirect}} = 6.86 \times 0.65 = \mathbf{4.46 \text{ kWh/kg}}$$

4. Evidence Base

Source	Finding	Relevance
Koç & Çinçik (2010), Table 3	Weaving plant: machines 36.3%, compressors 29.4%, HVAC 27.1%, lighting 3.4%, cleaning 3.8%; total 5.06 kWh/kg	Primary structural template
Hasanbeigi & Price (2015)	Spinning plants: machines 78%, humidification 16%, lighting 3%, compressors 3%	Context on subsystem importance
Confederation of Indian Industry (n.d.)	Compressed air: 10–40% of total plant electricity	Supports compressor share estimate
Sakti et al. (2021)	Factory case: HVAC + chillers = 80% of total electricity	Upper-bound context
Çelik et al. (2025)	Cleaning/needle- bed maintenance: ~40% of machine stoppage time	Supports auxiliary allocation
ADEME & Ministère de la Transition Écologique (2024)	Ecobalyse upper estimate for knitting machine electricity: 2.4 kWh/kg	Anchor for derivation

5. Implementation in ecoinvent / Brightway

In each knitting dataset, direct and indirect electricity are aggregated into a **single exchange**:

Field	Entry
ecoinvent activity	`market group for electricity, medium voltage {GLO} \
Amount	E_machine(dtex) + 4.46 kWh/kg (circular) or E_direct + 4.46 kWh/kg (others)
Unit	kWh

Direct and indirect electricity are **not separated** as two exchanges because knitting plants rarely sub-meter machine vs. auxiliary loads. The split is documented here for transparency and can be implemented as two separate exchanges when primary sub-metered data become available.

6. Limitations and Update Protocol

1. The 4.46 kWh/kg value is a **temporary reference** derived from a weaving analogy. It should be replaced with knitting-specific sub-metered data when available.
2. The subsystem shares (35/25/30/6.5/3.5) are **editable**. They provide a transparent starting point and can be overridden per facility.
3. The derivation uses the **upper** EcoBalyse estimate (2.4 kWh/kg) as the anchor — this is intentionally conservative.
4. The value applies uniformly across all knitting technologies. If factory data suggest significantly different indirect loads for a specific technology (e.g., 3D knitting with robotic automation), a technology-specific value can be introduced.

7. References

- ADEME, & Ministère de la Transition Écologique. (2024). *Ecobalyse methodological guide — Textiles and footwear* (Version 1.0). Paris, France.
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